AWS Secrets Manager — Important Interview Questions & Answers

1.What is AWS Secrets Manager?
AWS Secrets Manager is an AWS service used to securely store, manage, and retrieve sensitive information such as database passwords, API keys, and other secrets.
Simple ga gurthupettuko:
Secrets Manager ante passwords, API keys and sensitive information ni secure ga store chese AWS service.
Eng: AWS Secrets Manager securely stores and manages sensitive information.

2.What is a Secret?
A Secret is sensitive information stored securely in AWS Secrets Manager. It can contain database credentials, API keys, passwords, or other confidential information.
Simple ga gurthupettuko:
Secret ante password, API key or sensitive information ni secure ga store chese data.
Eng: A Secret is sensitive information securely stored in Secrets Manager.

3.Why do we use AWS Secrets Manager?
We use Secrets Manager to securely store and manage sensitive credentials instead of hardcoding them in application code or configuration files.
Simple ga gurthupettuko:
Passwords and credentials ni code lo hardcode cheyyakunda secure ga store cheyyadaniki Secrets Manager use chestham.
Eng: Secrets Manager is used to securely store credentials without hardcoding them in applications.

4.What is Secret Rotation?
Secret Rotation is the process of regularly changing a stored secret, such as a database password, to improve security. AWS Secrets Manager can automate rotation for supported secrets and configurations.
Simple ga gurthupettuko:
Secret Rotation ante security kosam password or secret ni regular ga change cheyyadam.
Eng: Secret Rotation regularly changes secrets to improve security.

5.Why do we use Secret Rotation?
We use Secret Rotation to reduce the risk of long-term use of the same password or credential. Regularly changing secrets improves security.
Simple ga gurthupettuko:
Same password long time use cheyyakunda security improve cheyyadaniki rotation use chestham.
Eng: Secret Rotation reduces the risk of using the same credentials for a long time.

6.What is Encryption in Secrets Manager?
Encryption protects the secret value by storing it in an encrypted form. AWS Secrets Manager uses AWS KMS keys to encrypt secret values.
Simple ga gurthupettuko:
Encryption ante Secrets Manager lo stored secret value ni secure encrypted form lo maintain cheyyadam.
Eng: Secrets Manager encrypts secret values using AWS KMS.

7.What are Secret Policies?
Secret Policies are resource-based policies that control access to a secret. They can define which IAM principals or AWS accounts can access the secret.
Simple ga gurthupettuko:
Secret Policy ante secret ni evaru access cheyyalo control chese policy.
Eng: A Secret Policy controls who can access a secret.

8.What is the difference between Secrets Manager and IAM?
Secrets Manager is used to securely store sensitive information such as passwords and API keys, while IAM is used to manage identities and permissions for AWS resources.
Simple ga gurthupettuko:
Secrets Manager secrets store cheyyadaniki, IAM users, roles and permissions manage cheyyadaniki use chestham.
Eng: Secrets Manager stores secrets, while IAM manages identities and permissions.

9.What is the difference between Secrets Manager and Parameter Store?
Secrets Manager is designed for securely storing and managing secrets and supports features such as automatic rotation. Parameter Store is used to store configuration values and parameters and can also store secure values.
Simple ga gurthupettuko:
Secrets Manager mainly secrets kosam, Parameter Store mainly configuration values and parameters kosam use chestham.
Eng: Secrets Manager manages secrets, while Parameter Store stores parameters and configuration values.

10.Can an application retrieve a secret from Secrets Manager?
Yes. An application can retrieve a secret using the AWS SDK, CLI, or supported AWS integrations. The application must have the required IAM permissions.
Simple ga gurthupettuko:
Application AWS SDK or CLI use chesi secret retrieve cheyyachu, required IAM permission unte.
Eng: Applications can retrieve secrets using AWS tools when they have the required permissions.

11.How would you allow an EC2 instance to access a secret?
I would attach an IAM Role to the EC2 instance and grant the required Secrets Manager permissions through an IAM policy. The secret policy must also allow access when applicable.
Simple ga gurthupettuko:
EC2 ki IAM Role attach chesi Secrets Manager secret access permission ivvachu.
Eng: I would use an IAM Role with the required permissions to allow EC2 to access a secret.

12.How would you secure database passwords in AWS?
I would store the database username and password in AWS Secrets Manager instead of keeping them in application code or configuration files. I would also use IAM permissions, encryption, and secret rotation where appropriate.
Simple ga gurthupettuko:
Database password ni code lo store cheyyakunda Secrets Manager lo secure ga store chestham.
Eng: I would store database credentials in Secrets Manager instead of hardcoding them.

13.What happens when a secret is rotated?
When a secret is rotated, the stored secret value is changed according to the configured rotation process. Applications that retrieve the secret dynamically can use the updated value.
Simple ga gurthupettuko:
Secret rotate ayite old value change ayi new secret value update avutundi.
Eng: When a secret is rotated, its stored value is updated.

14.Can Secrets Manager use KMS?
Yes. AWS Secrets Manager uses AWS KMS for encryption of secret values. Customer managed KMS keys can also be used when required.
Simple ga gurthupettuko:
Secrets Manager secret values ni encrypt cheyyadaniki KMS use chestundi.
Eng: Secrets Manager uses AWS KMS to encrypt secret values.

15.What happens if an application does not have permission to access a secret?
The application will receive an access denied error when it tries to retrieve the secret. We should check the IAM permissions and the secret resource policy if applicable.
Simple ga gurthupettuko:
Permission lekapothe secret retrieve chesetappudu Access Denied vastundi.
Eng: Without the required permissions, the application receives an Access Denied error.

16.How would you troubleshoot a Secrets Manager Access Denied error?
I would check the IAM Role or User permissions, secret resource policy, KMS permissions if applicable, secret ARN, and whether the correct AWS account and Region are being used.
Simple ga gurthupettuko:
Access Denied vaste IAM → Secret Policy → KMS Permission → Secret ARN → Region check chestham.
Eng: I would check IAM permissions, secret policies, KMS permissions, the secret ARN, and the Region.

17.Why should we avoid storing passwords in application code?
Passwords stored in application code can be exposed through source code repositories, logs, configuration files, or unauthorized access. Secrets Manager provides a more secure way to manage sensitive credentials.
Simple ga gurthupettuko:
Passwords code lo unte leak ayye chance untundi, anduke Secrets Manager use chestham.
Eng: We avoid hardcoding passwords because they can be exposed through application code and repositories.

18.What is the most important thing to remember about AWS Secrets Manager?
AWS Secrets Manager is used to securely store, manage, encrypt, retrieve, and rotate sensitive information such as passwords and API keys. The important concepts are Secrets, Rotation, Encryption, and Secret Policies.
Simple ga gurthupettuko:
Secrets Manager ante passwords and sensitive information ni secure ga store, manage, and rotate cheyyadaniki use chese AWS service.
Eng: AWS Secrets Manager securely stores, manages, encrypts, and rotates sensitive information.